AI/ML Task-2 Report

Project Title: Feature Engineering, Model Optimization & Performance Comparison

This project focuses on applying feature engineering and machine learning techniques on the California Housing Dataset. Multiple regression models were trained and evaluated to compare their predictive performance.

Methodology

The California Housing Dataset was imported using scikit-learn. The dataset was inspected for missing values and feature consistency. Feature scaling was performed using StandardScaler to normalize the feature values and improve model performance. The dataset was divided into training and testing sets using an 80:20 ratio.

Three regression models were implemented:
- Linear Regression
- Ridge Regression
- Decision Tree Regressor

Each model was trained on the training data and evaluated on the testing data using RMSE (Root Mean Squared Error) and R^2 Score metrics.

Results

The performance of all models was compared using evaluation metrics. Linear Regression provided baseline performance, while Ridge Regression improved generalization by reducing overfitting through regularization.

The Decision Tree Regressor achieved the best predictive performance because it effectively captured non-linear relationships present in the dataset.

Model	**Performance Summary**
Linear Regression	Simple baseline model with good interpretability
Ridge Regression	Reduced overfitting using regularization
Decision Tree Regressor	Best accuracy and handled non-linear patterns effectively

Conclusion

The project successfully demonstrated feature engineering, preprocessing, model training, and model evaluation techniques. Feature scaling improved model efficiency, while evaluation metrics helped identify the best-performing model.

Among all implemented models, the Decision Tree Regressor was selected as the best model because it achieved superior predictive accuracy and captured complex data relationships more effectively than linear models.

This task enhanced understanding of regression models, feature engineering, and machine learning workflow implementation using Python and scikit-learn.